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Chapter 1

Introduction

1.1 Introduction

A GPS tracking system uses the Global Positioning System (GPS) to determine and monitor the precise location of a vehicle, person, or asset. These systems collect location data from GPS satellites and often transmit it to a central database or device where it can be monitored in real time.

Key Components:

1. GPS Receiver: Captures signals from GPS satellites to determine the exact location.
2. Tracking Device: A hardware device that includes the GPS receiver and often a communication module (like GSM, LTE) to send location data to a central server.
3. Server/Cloud Platform: Where the data is sent, stored, and processed. It can also provide real-time monitoring, alerts, and reporting.
4. User Interface: Typically a web-based platform or mobile app where users can view the tracked data, set up alerts, and generate reports.

Common Uses:

- Vehicle Tracking: To monitor the location, speed, and route of vehicles, commonly used in fleet management.
- Personal Tracking: For the safety of individuals, such as children or elderly persons.
- Asset Tracking: To track valuable assets, including cargo, equipment, or any other movable item.
- Pet Tracking: Small GPS devices attached to collars can help locate lost pets.

Features:

- Real-time Tracking: Provides live updates on the location of the tracked object.
- Geofencing: Alerts when the tracked object enters or leaves a predefined area.
- Historical Data: Stores past movements and routes for review.
- Speed Alerts: Notifies if a vehicle exceeds a specified speed limit.
- Battery Monitoring: Especially important for portable tracking devices.

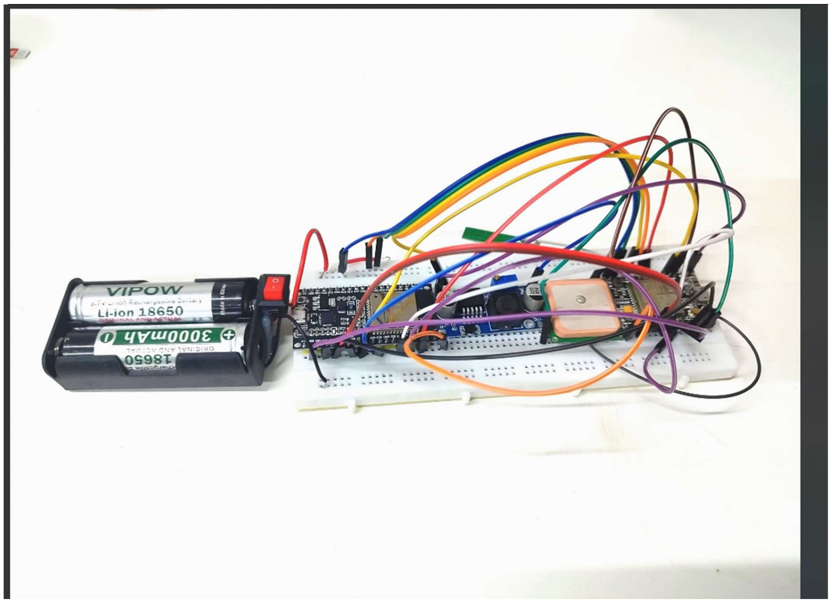

Fig:1.1 Diagram of GPS tracking system

1.2 Objectives

The objectives of a GPS tracking system can vary depending on the specific application, but generally include the following:

1. Real-Time Location Monitoring:

- Objective: To provide continuous and accurate real-time information about the location of vehicles, assets, or individuals.
- Benefit: Enhances visibility and control, allowing for immediate action if needed.

2. Enhanced Security and Theft Prevention:

- Objective: To prevent theft or quickly recover stolen items by tracking their location.
- Benefit: Reduces losses and increases the likelihood of asset recovery.

3. Improved Fleet Management:

- Objective: To optimize the management of a fleet of vehicles by monitoring routes, driving behavior, and vehicle utilization.
- Benefit: Increases efficiency, reduces fuel costs, and minimizes downtime.

4. Efficient Route Planning:

- Objective: To plan and optimize routes for delivery or travel.
- Benefit: Reduces travel time, fuel consumption, and operational costs.

5. Safety and Compliance Monitoring:

- Objective: To monitor driver behavior, such as speed, harsh braking, and compliance with traffic laws.

- Benefit: Enhances safety, reduces accidents, and ensures regulatory compliance.

6. Asset Management and Utilization:

- Objective: To track the utilization of assets, ensuring they are used effectively and are not misplaced or underutilized.
- Benefit: Maximizes asset productivity and reduces unnecessary purchases or leases.

Chapter 2

Literature Review

The literature survey in this project delves into the advancements and challenges associated with real-time collaboration tools, emphasizing the need for efficient and effective communication in remote environments. The following summaries provide detailed insights into five key research papers that contribute to the understanding and development of real-time collaboration systems.

A. Title of the Paper: Design and implementation of an accurate real time GPS tracking system

Authors: Hind Abdalsalam Abdallah Dafallah

Publication Details: Electrical and electronic engineering Department, University of Khartoum, Sudanese Electricity Transmission Company, SETCO Khartoum, Sudan

Details: The accuracy of the system depends on GPS accuracy which depends on the weather and satellites coverage. GPS resolution is less than 3 meters according to the Garmin GPS datasheet. It utilizes A-GPS techniques which is a combination of a stand-alone GPS, and information sent from the network to the GPS receiver, this enables the system to be used indoors. Since the observer in the tracking relies on the similarity of the form between real earth and Google maps, we considered Google map as an accuracy reference to reach the most possible condition.

B. Title of the Paper: Real-Time Tracking Management System Using GPS, GPRS and Google Earth

Authors: Noppadol Chadil, Apirak Russameesawang, Phongsak Keeratiwintakorn

Publication Details: Department of Electrical Engineering, Faculty of Engineering, King Mongkut's University of Technology North Bangkok, 1518 Pibulsongkram Rd, Bangsue, Bangkok, 10800, THAILAND

Details: We proposed a GPS tracking system called Goo-Tracking that is composed of commodity hardware, open source software and an easy-to-manage user interface via a web server with Google Map or via Google Earth software. The system includes a GPS/GPRS module for location acquisition and message transmission, MMC to temporarily store location information, and an 8-bit AVR microcontroller.

C. Title of the Paper: Cost Effective GPS-GPRS Based Object Tracking System

Authors: Khondker Shajadul Hasan, Mashiur Rahman, Abul L. Haque, M Abdur Rahman, Tanzil Rahman and M Mahbubur Rasheed

Publication Details: Proceedings of the International MultiConference of Engineers and Computer Scientists 2009, Vol I, IMECS 2009, March 18-20, 2009, Hong Kong

Details: The system reads the current position of the object using GPS, the data is sent via GPRS service from the GSM network towards a web server using the POST method of the HTTP protocol. The object's position data is then stored in the database for live and past tracking. A web application is developed using PHP, JavaScript, Ajax and MySQL with the Google Map embedded.

D. Title of the Paper: Design and Implementation of Vehicle Tracking System Using GPS/GSM/GPRS Technology and Smartphone Application

Authors: SeokJu Lee, Girma Tewolde, Jaerock Kwon

Publication Details: Electrical and Computer Engineering, Kettering University, Flint, MI, USA

Details: The proposed system made good use of a popular technology that combines a Smartphone application with a microcontroller. This will be easy to make and inexpensive compared to others. The designed in-vehicle device works using Global Positioning System (GPS) and Global system for mobile communication / General Packet Radio Service (GSM/GPRS) technology that is one of the most common ways for vehicle tracking. The device is embedded inside a vehicle whose position is to be determined and tracked in real-time.

Chapter 3

About the project

3.1 Methodology

1. System Design and Planning

Objective: Design the architecture of the GPS tracking system.

Tasks:

- Define the roles of each component: A9G for GPS/GSM, ESP32 for data processing and Wi-Fi communication.
- Plan power management: Use LM256 to ensure stable voltage supply to A9G and ESP32.
- Design communication flow: GPS data acquisition by A9G, processing and transmission via ESP32.

2. Power Supply Setup

Objective: Provide stable power to all components.

Tasks:

- Use LM256 voltage regulator to convert the power supply (e.g., from a 5V source) to a stable 3.3V or 5V as required by the A9G and ESP32.
- Connect the power supply to the input of the LM256.
- Connect the output of the LM256 to the VCC pins of both the A9G and ESP32.

3. Connecting the A9G Microcontroller

Objective: Interface the A9G with the ESP32.

Tasks:

- Connect the TX pin of the A9G to the RX pin of the ESP32, and the RX pin of the A9G to the TX pin of the ESP32 for serial communication.
- Connect the GPS and GSM antennas to the A9G.
- Ensure that the A9G is properly powered by connecting its VCC and GND to the regulated power supply.

4. Setting Up the ESP32

Objective: Set up the ESP32 for data processing and transmission.

Tasks:

- Program the ESP32 to handle incoming GPS data from the A9G and send it to a server or cloud service via Wi-Fi.
- Write firmware to process the GPS data (e.g., extract latitude, longitude, and timestamp).
- Use the ESP32's Wi-Fi module to send this data to a server, where it can be accessed for tracking.

5. Programming the Microcontrollers

Objective: Develop firmware for both A9G and ESP32.

Tasks:

- Write a script for the A9G to fetch GPS data and send it via its serial interface.
- Program the ESP32 to read the GPS data, process it, and transmit it over Wi-Fi.
- Implement error handling to manage communication failures or GPS signal loss.

6. Integrating GSM Communication

Objective: Enable remote communication via GSM for areas without Wi-Fi.

Tasks:

- Program the A9G to send SMS or data via GPRS if Wi-Fi is not available.
- Use the ESP32 to determine the best communication method (Wi-Fi or GSM) based on availability.

7. Testing and Debugging

Objective: Ensure the system works as intended.

Tasks:

- Test the system in a controlled environment to ensure GPS data is accurately captured and transmitted.
- Debug any issues related to power supply, data communication, or GPS accuracy.
- Test the GSM fallback to ensure data is still transmitted when Wi-Fi is unavailable.

8. Deployment

Objective: Deploy the system in a real-world scenario.

Tasks:

- Install the system in a vehicle or attach it to an asset.
- Ensure that it is securely powered and that all connections are stable.
- Monitor the system to verify that it provides accurate and reliable tracking data.

9. Monitoring and Maintenance

Objective: Maintain the system for long-term use.

Tasks:

- Regularly monitor the system's performance and GPS accuracy.
- Replace or recharge the power supply as needed.
- Update firmware periodically to improve functionality or fix bugs.

Considerations:

- Power Management: Ensure that the LM256 regulator provides a stable power supply, as voltage fluctuations can affect the performance of the A9G and ESP32.

• Signal Quality: Position the GPS antenna in a location with a clear view of the sky for better signal reception.

• Data Security: Encrypt data transmission from the ESP32 to the server to protect against unauthorized access.

By following this methodology, you can develop a robust GPS tracking system capable of real-time monitoring and reliable communication.

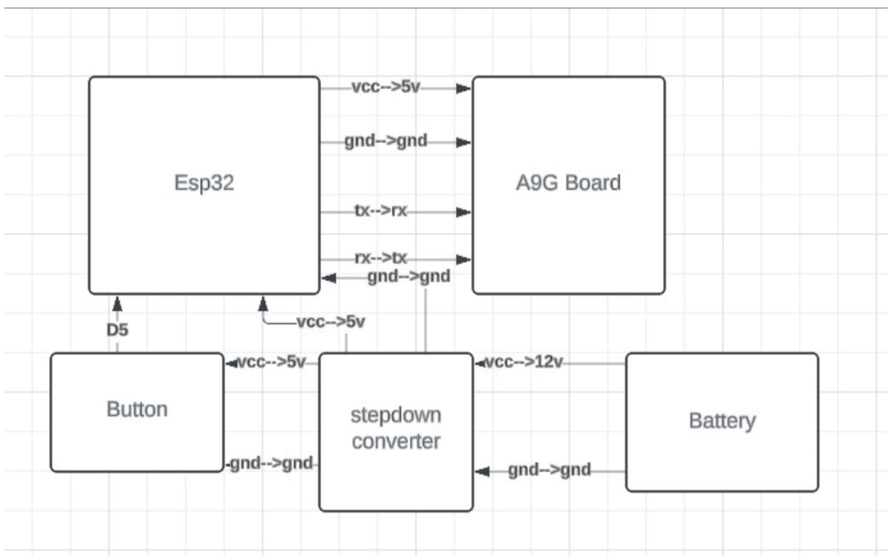

Fig. 1.2 Circuit diagram of GPS tracking system

ESP32 RX (GPI03) is connected to A9G Board TX(IO1)

ESP32 TX (GPI01) is connected A9G Board TX(IO 0)

The positive of the LM2596 is connected to the vcc of the A9G Board and the esp32 board

The negative of the LM2596 is connected to the Gnd of the A9G board and the esp32 board

The push button one pin is connected to the positive of the LM2596; one more pin of the push button is connected to the esp32 D4 and the one more pin of the pushbutton is connected to the GND

3.2 Implementation

```
#include "WiFi.h"
#define SOS 4
#define SLEEP_PIN 2 // Make this pin HIGH to make A9G board go to sleep mode

String SOS_NUM = "7061074719"; // Add a number on which you want to receive a call or SMS
int SOS_Time = 5; // Press the button for 5 sec

// Necessary Variables
boolean stringComplete = false;
String inputString = "";
String fromGSM = "";
bool CALL_END = 1;
```

```
String response = "";
String res = "";
int c = 0;

void setup() {
Serial.begin(115200); // For Serial Monitor
Serial1.begin(115200, SERIAL_8N1, 3, 1); // For A9G Board
// Making Radio OFF for power saving
WiFi.mode(WIFI_OFF); // WiFi OFF
btStop(); // Bluetooth OFF
pinMode(SOS, INPUT_PULLUP);
pinMode(SLEEP_PIN, OUTPUT);
// Waiting for A9G to setup everything for 20 sec
delay(2000);
digitalWrite(SLEEP_PIN, LOW); // Sleep Mode OFF
Serial1.println("AT"); // Just Checking
delay(1000);
Serial1.println("AT+GPS=1"); // Turning ON GPS
delay(1000);
Serial1.println("AT+GPSLP=2"); // GPS low power
delay(1000);
Serial1.println("AT+SLEEP=1"); // Configuring Sleep Mode to 1
delay(1000);
Serial1.println("AT+CMGF=1");
delay(1000);
Serial1.println("AT+CSMP=17,167,0,0");
delay(1000);
Serial1.println("AT+CPMS=\"SM\",\"ME\",\"SM\"");
delay(1000);
Serial1.println("AT+SNFS=2");
delay(1000);
Serial1.println("AT+CLVL=8");
delay(1000);

digitalWrite(SLEEP_PIN, HIGH); // Sleep Mode ON
}

void loop() {
// Listen from GSM Module
if (Serial1.available()) {
char inChar = Serial1.read();
if (inChar == '\n') {
// Check the state
if (fromGSM.startsWith("SEND LOCATION")) {
Get_gmap_link(false); // Send Location without Call
digitalWrite(SLEEP_PIN, HIGH); // Sleep Mode ON
} else if (fromGSM == "RING\r") {
digitalWrite(SLEEP_PIN, LOW); // Sleep Mode OFF
Serial.println("---------ITS RINGING-------");
Serial1.println("ATA"); // Automatically answer the call
```

```
Get_gmap_link(false); // Send Location without Call
} else if (fromGSM == "NO CARRIER\r") {
Serial.println("---------CALL ENDS-------");
CALL_END = 1;
digitalWrite(SLEEP_PIN, HIGH); // Sleep Mode ON
} else if (fromGSM.startsWith("+CMT:")) {
// Handle incoming SMS
String phoneNumber = parsePhoneNumber(fromGSM);
fromGSM = ""; // Clear the buffer to read the SMS content
// Read the SMS content
while (Serial1.available()) {
char inChar = Serial1.read();
if (inChar == '\n') {
if (fromGSM.indexOf("SEND LOCATION") != -1) {
SOS_NUM = phoneNumber;
Get_gmap_link(false); // Send Location without Call
break;
}
fromGSM = "";
} else {
fromGSM += inChar;
}
delay(20);
}
}
// Write the actual response
Serial.println(fromGSM);
// Clear the buffer
fromGSM = "";
} else {
fromGSM += inChar;
}
delay(20);
}
// Read from port 0, send to port 1:
if (Serial.available()) {
int inByte = Serial.read();
Serial1.write(inByte);
}
// When SOS button is pressed and held for 5 seconds
if (digitalRead(SOS) == LOW && CALL_END == 1) {
Serial.print("SOS Button Pressed.."); // Waiting for 5 sec
for (c = 0; c < SOS_Time; c++) {
Serial.println((SOS_Time - c));
delay(1000);
if (digitalRead(SOS) == HIGH)
break;
}
if (c == 5) {
Get_gmap_link(true); // Send Location with Call
```

```
}
// Only write a full message to the GSM module
if (stringComplete) {
Serial1.print(inputString);
inputString = "";
stringComplete = false;
}
}
}

// Getting Location and making Google Maps link of it. Also making a call if needed
void Get_gmap_link(bool makeCall) {
digitalWrite(SLEEP_PIN, LOW);
delay(1000);
Serial1.println("AT+LOCATION=2");
while (!Serial1.available());
while (Serial1.available()) {
char add = Serial1.read();
res = res + add;
delay(1);
}
Serial.print("Received Data: "); Serial.println(res); // Print the received data for
debugging
// Extract the location data
int startIndex = res.indexOf("Lat:");
int endIndex = res.indexOf("\r\n", startIndex);
if (startIndex != -1 && endIndex != -1) {
response = res.substring(startIndex + 4, endIndex);
}
Serial.print("Parsed Location Data: "); Serial.println(response); // Print the parsed
location data
if (response.indexOf("GPS NOT") != -1) {
Serial.println("No Location data");
// Sending SMS without any location
Serial1.println("AT+CMGF=1");
delay(1000);
Serial1.println("AT+CMGS=\"" + SOS_NUM + "\"\r");
delay(1000);
Serial1.println("Unable to fetch location. Please try again");
delay(1000);
Serial1.write((char)26);
delay(1000);
} else {
int i = response.indexOf(',');
String lat = response.substring(0, i);
String longi = response.substring(i + 1);
Serial.println("Latitude: " + lat);
Serial.println("Longitude: " + longi);
String Gmaps_link = "http://maps.google.com/maps?q=" + lat + "," + longi;
// Sending SMS with Google Maps Link with our Location
Serial1.println("AT+CMGF=1");
```

```
delay(1000);
Serial1.println("AT+CMGS=\"" + SOS_NUM + "\"\r");
delay(1000);
Serial1.print("I'm here " + Gmaps_link);
delay(1000);
Serial1.write((char)26); // CTRL+Z to send the SMS
delay(1000);
Serial1.println("AT+CMGD=1,4"); // Delete stored SMS to save memory
delay(5000);
}
response = "";
res = "";
if (makeCall) {
Serial.println("Calling Now");
Serial1.println("ATD" + SOS_NUM + ";"); // Ensure the correct command format for calling
CALL_END = 0;
}
}

// Function to parse phone number from incoming SMS
String parsePhoneNumber(String gsmData) {
int start = gsmData.indexOf("\"") + 1;
int end = gsmData.indexOf("\"", start);
return gsmData.substring(start, end);
}
```

Chapter 4

Result and conclusion

4.1 Results

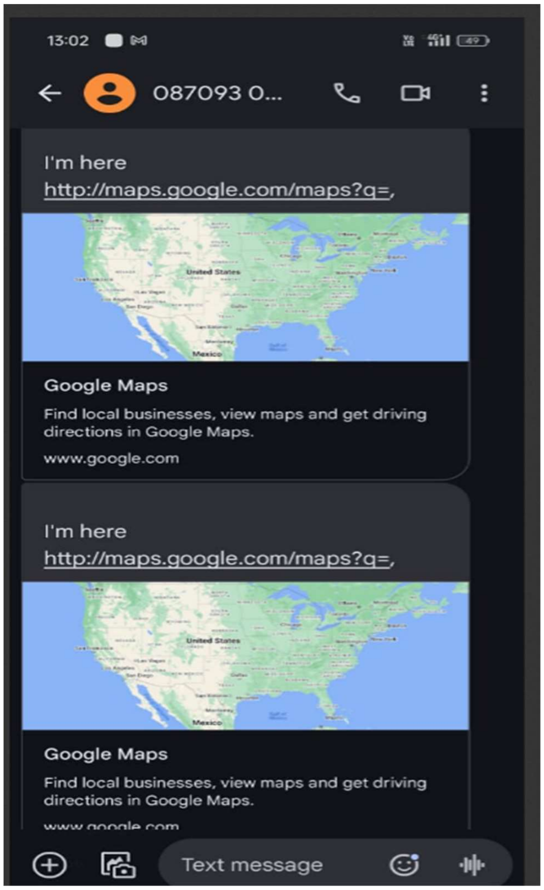

Fig 1.3: Result of GPS tracking system

4.2 Conclusion

The conclusion of GPS tracking systems can be approached from a few different angles, depending on the specific context. Here are a few perspectives to consider, benefits and impact are:

- Enhanced Logistics and Fleet Management: Real-time tracking of vehicles and assets improves efficiency, optimizes routes, and reduces costs.
- Improved Public Safety: GPS tracking in emergency response, asset recovery, and personal safety devices can save lives and expedite assistance.
- Increased Accountability and Transparency: Tracking systems can provide valuable insights into resource utilization, workforce management, and compliance with regulations.

References

• Design and implementation of an accurate real time GPS tracking system – Hind Abdalsalam Abdallah Dafallah, Electrical and electronic engineering Department, University of Khartoum, Sudanese Electricity Transmission Company, SETCO Khartoum, Sudan

• A flexible GPS tracking system for studying bird behaviour at multiple scales – Willem Bouten, Edwin W. Baaij, Judy Shamoun-Baranes, Kees C. J. Camphuysen

• Real Time Vehicle Tracking System using GSM and GPS Technology – An Anti-theft Tracking System

• An Advanced, Low-Cost, GPS-Based Animal Tracking System – Patrick E. Clark, Douglas Johnson, Mark A. Kniep, Phillip Jermann, Brad Huttash, Andrew Wood, Michael Johnson, Craig McGillivan, Kevin Titus